

4.2.2 Convert percentage to fraction and to decimal and vice versa.

Consider the following examples:

Example 1. Convert the given percentages into fraction and then into decimal.

(i) 15% (ii) 75%

Solution:

$$(i) \quad 15\% = \frac{15}{100} = \frac{\cancel{15}^3}{\cancel{100}^{20}} = \frac{3}{20}, \text{ it is fractional form.}$$

$$\text{and } 15\% = \frac{15}{100} = 0.15, \text{ it is decimal form.}$$

$$(ii) \quad 75\% = \frac{75}{100} = \frac{3 \times \cancel{25}^1}{4 \times \cancel{25}_1} = \frac{3}{4}, \text{ it is fractional form.}$$

$$\text{and } 75\% = \frac{75}{100} = 0.75, \text{ it is decimal form.}$$

Example 2. Convert the given decimals into fraction and then convert into percentage:

(i) 0.50 (ii) 2.45

Solution:

$$(i) \quad 0.50 = \frac{50}{100} = \frac{\cancel{50}^1}{\cancel{100}^2} = \frac{1}{2}$$

$$\text{Now, } 0.50 = \frac{50}{100} = 50\%$$

$$\text{Thus } 0.50 = \frac{1}{2} \text{ is in fractional form and } 0.50 = 50\%$$

$$(ii) \quad 2.45 = \frac{245}{100} = \frac{49 \times \cancel{5}^1}{20 \times \cancel{5}_1} = \frac{49}{20} = 2 \frac{9}{20} \text{ (fractional form)}$$

$$\text{Now, } 2.45 = \frac{245}{100} = 245\%$$

Example 3. Express the fraction $\frac{3}{5}$ as percentage.

Solution: For making denominator 100 multiply 3 and 5 by 20, we get the equivalent fraction.

$$\frac{3 \times 20}{5 \times 20} = \frac{60}{100}$$

This means 60%. Hence, $\frac{3}{5} = 60\%$

EXERCISE 4.14

A. Convert the given percentages into fraction and then into decimal.

- | | | | |
|---------|----------|-----------|-----------|
| (1) 25% | (2) 30% | (3) 35% | (4) 40% |
| (5) 65% | (6) 70% | (7) 80% | (8) 85% |
| (9) 95% | (10) 99% | (11) 105% | (12) 115% |

B. Convert the given decimals into fractions and then convert into percentage.

- | | | | |
|----------|-----------|-----------|------------|
| (1) 0.8 | (2) 0.34 | (3) 0.56 | (4) 0.63 |
| (5) 0.55 | (6) 0.66 | (7) 3.45 | (8) 3.60 |
| (9) 5.50 | (10) 2.05 | (11) 25.5 | (12) 55.25 |

C. Express the following fractions as percentages, giving your answer correct to 1 decimal place where necessary.

- | | | | | |
|--------------------|---------------------|---------------------|---------------------|---------------------|
| (1) $\frac{4}{5}$ | (2) $\frac{6}{25}$ | (3) $\frac{11}{20}$ | (4) $\frac{5}{8}$ | (5) $\frac{17}{40}$ |
| (6) $\frac{6}{12}$ | (7) $\frac{18}{60}$ | (8) $\frac{18}{30}$ | (9) $\frac{71}{50}$ | (10) $1\frac{4}{5}$ |

4.2.3 Solve real life problems involving percentages

Consider the following examples.

Example. There are 825 students studying in a school. 40 percent are girls. How many number of girls are studying in the school?

Solution:

Number of girls studying in the school = 40% of 825

$$= \frac{40}{100} \times 825 = 0.40 \times 825$$

$$= 330 \text{ girls}$$

Thus **330 girls** studying in the school.

EXERCISE 4.15

- (1) 480 students visited a book fair. Out of them 45 percent are boys. How many boys visited the book fair?
- (2) 825 students are studying in a school. 60 percent students were regular for whole of the session. How many students were regular in the school?
- (3) 900 employees are in a company. 70 percent employees were using English frequently. How many employees are using English frequently?
- (4) 65 percent of the houses in the colony have computers. There are 2450 houses in all. How many houses have computers?
- (5) In a school total number of students are 1200. In class V have 20% of the students. How many students in class V?
- (6) There are 450 cars are in a car parking and 20% of these are white in colour. What is the number of white cars?
- (7) The original price of a dress is Rs 1650. During a sale, it is sold at 9% discount. What is the price of a dress?

REVIEW EXERCISE 4

A. Fill in the blanks.

- (1) $42.0 \div \underline{\hspace{2cm}} = 0.42$
- (2) $12.04 \times \underline{\hspace{2cm}} = 1204.$
- (3) In decimals $65\% = \underline{\hspace{2cm}}.$
- (4) $4 + 0.4 + 0.04 + 0.004 = \underline{\hspace{2cm}}.$
- (5) When we round off 4.956 to two decimal places, we get $\underline{\hspace{2cm}}.$

B. Solve:

- (1) Arrange 0.016, 2.087, 1.995, 0.463 in ascending order.
- (2) Solve: $9.123 - 5.865 + 2.403$
- (3) Height of an electric pole is 14.95 m and height of mobile tower is 25.04 m. How much higher is the mobile tower than electric pole?
- (4) Verify: $4.5 \times 3.2 = 45 \times 0.32$
- (5) Divide 0.72 by 1.2

C. Solve the problems.

- (1) 3.6 m cloth is required to make a dress. How many such dresses can be prepared from 14.4 m of cloth?
- (2) Aftab gets 84% marks in all subjects of computer science. If the total marks are 2000. Find the marks he obtained.
- (3) Find the value of:
(a) 3.5% of 450 (b) 0.45% of 760
(c) 110% of 220 (c) 1.1% of 1000

DISTANCE, TIME AND TEMPERATURE

5.1 DISTANCE

5.1.1 Convert kilometres to metres and metres to kilometres

We know the units of measuring length. Following is the table showing relationship among units of length.

$$10 \text{ millimetres (mm)} = 1 \text{ centimetre (cm)}$$

$$100 \text{ centimetres (cm)} = 1 \text{ metre (m)}$$

$$1000 \text{ metres (m)} = 1 \text{ kilometre (km)}$$

Units of length can be converted.

It should be noted that when the bigger unit of lengths are converted into smaller units, we multiply the bigger units with their equivalent smaller units.

Similarly, when the smaller units are converted into bigger units, we divide the smaller units by the equivalent bigger units.

We already learnt to convert kilometres to metres, we multiply the number of kilometres by 1000.

Examples:

- (1) Convert 5 km to metres (2) Convert 8 km 150 m to metres

Solution:

$$\begin{aligned}\text{Distance} &= 5 \text{ km} \\ &= (5 \times 1000) \text{ m} \\ &= 5000 \text{ m}\end{aligned}$$

Solution:

$$\begin{aligned}\text{Distance} &= 8 \text{ km } 150 \text{ m} \\ &= (8 \times 1000) \text{ m} + 150 \text{ m} \\ &= 8000 \text{ m} + 150 \text{ m} \\ &= 8150 \text{ m}\end{aligned}$$

To convert metres to kilometres, we divide the number of metres by 1000.

Teacher's Note

Teacher should explain the units of length and time. He/she should also revise the procedure of conversion into different units.

Examples.**(1) Convert 12000 m to km****Solution:**

$$\begin{aligned} 12000 \text{ m} &= (12000 \div 1000) \text{ km} \\ &= \frac{12000}{1000} \text{ km} = \frac{12}{1} \text{ km} = 12 \text{ km} \end{aligned}$$

(2) Convert 18450 m to km and m**Solution:**

$$\begin{aligned} 18450 \text{ m} &= 18000 \text{ m} + 450 \text{ m} \\ &= \frac{18000}{1000} \text{ km} + 450 \text{ m} \\ &= 18 \text{ km } 450 \text{ m} \end{aligned}$$

$$\begin{array}{r} 18 \text{ km} \\ 1000 \overline{) 18450} \\ \underline{- 1000} \\ 8450 \\ \underline{8000} \\ 450 \text{ m} \end{array}$$

5.1.2 Convert metres to centimetres and centimetres to metres

We already learnt to convert metres to centimetres in previous class. We multiply the numbers of metres by 100.

Examples:**(1) Convert 11 m to cm****Solution:**

$$\begin{aligned} 11 \text{ m} &= (11 \times 100) \text{ cm} \\ &= 1100 \text{ cm} \end{aligned}$$

(2) Convert 15 m 30 cm to cm**Solution:**

$$\begin{aligned} &15 \text{ m } 30 \text{ cm} \\ &= (15 \times 100) \text{ cm} + 30 \text{ cm} \\ &= 1500 \text{ cm} + 30 \text{ cm} \\ &= 1530 \text{ cm} \end{aligned}$$

To convert centimetres to metres, we divide the number of centimetres by 100.

Examples:**(1) Convert 1400 cm to m****Solution:**

$$\begin{aligned}
 1400 \text{ cm} &= (1400 \div 100) \text{ m} \\
 &= \overset{14}{\cancel{1400}} \times \frac{1}{\underset{1}{\cancel{100}}} \text{ m} = 14 \text{ m}
 \end{aligned}$$

(2) Convert 2436 cm to m and cm**Solution:**

$$\begin{aligned}
 2436 \text{ cm} &= 2400 \text{ cm} + 36 \text{ cm} \\
 &= (2400 \div 100) \text{ m} + 36 \text{ cm} \\
 &= \left(\overset{24}{\cancel{2400}} \times \frac{1}{\underset{1}{\cancel{100}}} \right) \text{ m} + 36 \text{ cm} = 24 \text{ m } 36 \text{ cm}
 \end{aligned}$$

5.1.3 Convert centimetres to millimetres and millimetres to centimetres

We have already learnt about conversion of centimetres to millimetres, we multiply the number of centimetres by 10.

Examples.**(1) Convert 16 cm to mm****Solution:**

$$\begin{aligned}
 16 \text{ cm} &= (16 \times 10) \text{ mm} \\
 &= 160 \text{ mm}
 \end{aligned}$$

(2) Convert 25 cm 4 mm to mm**Solution:**

$$\begin{aligned}
 &25 \text{ cm } 4 \text{ mm} \\
 &= 25 \text{ cm} + 4 \text{ mm} \\
 &= (25 \times 10) \text{ mm} + 4 \text{ mm} \\
 &= 250 \text{ mm} + 4 \text{ mm} = 254 \text{ mm}
 \end{aligned}$$

To convert millimetres to centimetres, we divide the number of millimetres by 10.

Examples.

(3) Convert 350 mm to cm

Solution:

$$350 \text{ mm} = (350 \div 10) \text{ cm}$$

$$\left(\cancel{350} \times \frac{1}{\cancel{10}} \right) \text{ cm} = 35 \text{ cm}$$

(4) Convert 145 mm to cm and mm

Solution:

$$145 \text{ mm}$$

$$= (145 \div 10) \text{ cm}$$

$$= 14.5 \text{ cm}$$

$$= 14 \text{ cm } 5 \text{ mm}$$



Activity

Convert the following:

- (1) $9 \text{ km} = 9 \times 1000 = 9000 \text{ m}$
- (2) $2500 \text{ m} = 2500 \div 1000 = 2.5 \text{ km}$
- (3) $3784 \text{ m} = \underline{\hspace{2cm}} \text{ km}$
- (4) $3000 \text{ m} = \underline{\hspace{2cm}} \text{ km}$
- (5) $24 \text{ m} = \underline{\hspace{2cm}} \text{ cm}$
- (6) $350 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$
- (7) $200 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$
- (8) $4 \text{ m } 58 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$
- (9) $3 \text{ km } 400 \text{ m} = \underline{\hspace{2cm}} \text{ m}$
- (10) $1320 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$
- (11) $425 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$
- (12) $250 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$
- (13) $500 \text{ mm} = \underline{\hspace{2cm}} \text{ m}$
- (14) $10 \text{ mm} = \underline{\hspace{2cm}} \text{ cm}$
- (15) $28 \text{ cm } 5 \text{ mm} = \underline{\hspace{2cm}} \text{ mm}$

EXERCISE 5.1

A. How many kilometres and metres?

- (1) 1600 m (2) 2483 m (3) 1386 m
(4) 6034 m (5) 8324 m (6) 7945 m

B. Convert the lengths in metres and centimetres.

- (1) 400 cm (2) 750 cm (3) 385 cm (4) 810 cm
(5) 205 cm (6) 567 cm (7) 684 cm (8) 998 cm

C. Convert into centimetres and millimetres.

- (1) 35 mm (2) 634 mm (3) 593 mm (4) 400 mm
(5) 295 mm (6) 447 mm (7) 609 mm (8) 899 mm

D. Convert the following:

- (1) 8 km to m (2) 20 km 340 m to m
(3) 15 m to cm (4) 25 m 45 cm to m
(5) 35 cm to mm (6) 1200 m to km and m
(7) 3785 m to km and m (8) 1520 cm to m and cm
(9) 850 mm to cm (10) 4725 mm to cm and mm

E. How many metres?

- (1) 7000 mm (2) 8000 mm (3) 9000 mm
(4) 4000 cm (5) 1000 cm (6) 5000 cm
(7) 6000 mm (8) 10000 cm

5.1.4 Add and subtract measure of distances

Example 1. Add 3 m 35 cm distance to 8 m 75 cm.**Solution:**

m	cm
① 3	① 35
+ 8	75
12	10

Therefore addition of distances:
 $3 \text{ m } 35 \text{ cm} + 8 \text{ m } 75 \text{ cm} = 12 \text{ m } 10 \text{ cm}$

Example 2: Subtract 4 km 80 m from 11 km 15 m distances**Solution:**

km	m
¹⁰ 11	① 15
- 4	80
6	935

Therefore difference of distances:
 $11 \text{ km } 15 \text{ m} - 4 \text{ km } 80 \text{ m} = 6 \text{ km } 935 \text{ m}$

EXERCISE 5.2**A. Solve.**

- 43 km 15 m + 66 km 57 m
- 428 m 15 cm + 257 m 29 cm
- 5 km 860 m + 2 km 220 m
- 8 m 70 cm + 5 m 60 cm
- 3 km 918 m + 1 km 324 m
- 45 cm 8 mm + 65 cm 7 mm
- 8 km 750 m + 7 km 430 m
- 9 m 58 cm + 4 m 64 cm

B. Subtract.

- 12 km 75 m - 3 km 84 m
- 143 m 62 cm - 87 m 59 cm
- 8 km 546 m - 6 km 804 m
- 51 cm 3 mm - 27 cm 8 mm
- 5 km 150 m - 2 km 730 m
- 71 m 22 cm - 48 m 85 cm
- 7 km 505 m - 4 km 700 m
- 23 cm 2 mm - 17 cm 9 mm

5.1.5 Solve real life problems involving conversion, addition and subtraction of units of distance

Example 1.

A carpenter needs 2 pieces of wood of 2 m 30 cm long and 70 cm long. What is the total length of the pieces they need?

Solution:

	m	cm	
	① 2	30	
	+ 0	70	
	3 00		
Total length	3	00	(Would required is 3 m)

Example 2.

The length of an iron rod is 9 m 80 cm. A piece of 5 m 85 cm has been cut off from it. How much length of the rod is left?

Solution: Subtract 5 m 85 cm from 9 m 80 cm means
 $9 \text{ m } 80 \text{ cm} - 5 \text{ m } 85 \text{ cm}$.

	m	cm
Length of iron rod	9	80
Piece used	- 5	85
	3 95	

Length of iron rod left is **3 m 95 cm**

EXERCISE 5.3

Solve.

- (1) The length of a ribbon is 6 m 80 cm. How much ribbon is left, if 2m 88 cm has been cut off?
- (2) Jameel covered a distance of 589 m from his house to Jamia Masjid and then 868 m from Jamia Masjid to School. Find the total distance covered by him.
- (3) A car is 1 m 62 cm wide. A garage is 2m 41 cm wide. How much space is left when the car is in the garage?

- (4) Maria's home is distance as 375 m to the school. Railway station is 504 m far from her home. Which is farther from her home and how much?
- (5) The red part of a colour pencil is 65 mm long. The blue part is 57 mm long. What is the length of pencil? What is the answer in millimetres and centimetres?
- (6) In a walking race, in an specified time Hamid walked 2 km 102 m, Hussain walked only 1 km 985 m. How far ahead of Hussain was Hamid?
- (7) In a 250 kilometre car race, a car get accident at a distance of 134 km from the winning point. What distance had the car drive before accident?
- (8) Naeem is 142 cm tall. His friend is 8 cm taller than Naeem. How tall is his friend? Give the answer in metres.

5.2 TIME

5.2.1 Convert hours to minutes, minutes to seconds and vice versa

We already know the following relationship among various units of time.

$$60 \text{ seconds} = 1 \text{ minute}$$

$$60 \text{ minutes} = 1 \text{ hour}$$



To convert hours to minutes, we multiply the number of hours by 60.

Example: Convert 3 hours 20 minutes to minutes.

Solution: 3 hours 20 minutes

$$\begin{aligned} &= 3 \text{ hours} + 20 \text{ minutes} = (3 \times 60) \text{ minutes} + 20 \text{ minutes} \\ &180 \text{ minutes} + 20 \text{ minutes} \\ &= 200 \text{ minutes} \end{aligned}$$

To convert minutes into hours we divide the number of minutes by 60.

Examples:**(1) Convert 600 minutes to hours****Solution:**

$$\begin{aligned}
 &600 \text{ minutes} \\
 &600 \div 60 \text{ hours} \\
 &= 10 \text{ hours}
 \end{aligned}$$

(2) Convert 400 minutes to hours and minutes**Solution:**

$$\begin{aligned}
 &400 \text{ minutes} \\
 &= (400 \div 60) \text{ hours} \\
 &= 6 \text{ hours } 40 \text{ minutes}
 \end{aligned}$$

$$\begin{array}{r}
 6 \\
 60 \overline{) 400} \\
 \underline{- 360} \\
 40
 \end{array}$$

To convert minutes to seconds, we multiply the number of minutes by 60.

Examples:**(1) Convert 5 minutes 30 seconds to seconds****Solution:** Given:

$$\begin{aligned}
 &5 \text{ minutes } 30 \text{ seconds} \\
 &= 5 \text{ minutes} + 30 \text{ seconds} \\
 &= 5 \times 60 \text{ seconds} + 30 \text{ seconds} \\
 &= 300 \text{ seconds} + 30 \text{ seconds} \\
 &= 330 \text{ seconds}
 \end{aligned}$$

(2) Convert 12 minutes 45 seconds to seconds**Solution:** Given:

$$\begin{aligned}
 &12 \text{ minutes } 45 \text{ seconds} \\
 &= 12 \text{ minutes} + 45 \text{ seconds} \\
 &= 12 \times 60 \text{ seconds} + 45 \text{ seconds} \\
 &= 720 \text{ seconds} + 45 \text{ seconds} \\
 &= 765 \text{ seconds}
 \end{aligned}$$

To convert seconds to minutes, we divide the number of seconds by 60.

Example. Convert 660 seconds into minutes**Solution:** Given 660 seconds

$$\begin{aligned}
 &= (660 \div 60) \text{ minutes.} \\
 &= 11 \text{ minutes}
 \end{aligned}$$

**Activity****Convert 2 days into hours, minutes and seconds.**

$$2 \text{ days} = (2 \times 24) \text{ hours} = \underline{\hspace{2cm}} \text{ hours}$$

$$48 \text{ hour} = 48 \times 60 \text{ minutes} = \underline{\hspace{2cm}} \text{ minutes}$$

$$2880 \text{ minutes} = 2880 \times 60 \text{ seconds} = \underline{\hspace{2cm}} \text{ seconds}$$

EXERCISE 5.4**A. Convert into minutes.**

- | | |
|-------------------|--------------------|
| (1) 2 hours | (2) 8 hours |
| (3) 12 hours | (4) 1 day 3 hours |
| (5) 1 day 6 hours | (6) 1 day 10 hours |

B. Convert into seconds.

- | | |
|-----------------------|-----------------------|
| (1) 5 minutes | (2) 10 minutes |
| (3) 20 minutes | (4) 45 minutes |
| (5) 55 minutes | (6) 1 hour |
| (7) 1 hour 15 minutes | (8) 1 hour 25 minutes |

C. Convert into hours and minutes.

- | | |
|-------------------|-------------------|
| (1) 1,180 minutes | (2) 1,250 minutes |
| (3) 1,490 minutes | (4) 2,225 minutes |
| (5) 1,815 minutes | (6) 2,375 minutes |

D. Convert into minutes and seconds.

- | | |
|-------------------|-------------------|
| (1) 2,185 seconds | (2) 275 seconds |
| (3) 350 seconds | (4) 710 seconds |
| (5) 990 seconds | (6) 1,395 seconds |

E. Convert into hours, minutes and seconds.

- | | |
|-------------------|-------------------|
| (1) 3,800 seconds | (2) 4,100 seconds |
| (3) 4,360 seconds | (4) 4,595 seconds |
| (5) 4,725 seconds | (6) 4,915 seconds |

5.2.2 Addition and subtraction of units of time with carrying/borrowing

Example 1. Add 3 hours 55 minutes and 5 hours 40 minutes.**Solution:** We add 3 hours 55 minutes and 5 hours 40 minutes.

h	m	
① 3	55	55 minute + 40 minute = 95 minutes = 1 hour 35 minute
+ 5	40	Write 35 minutes under minutes column and carry 1 to hours column.
9	35	Thus, 9 hours 35 minutes

Example 2. Subtract 58 minutes 15 seconds from 72 minutes 30 seconds**Solution:** We subtract 58 minutes 15 seconds from 72 minutes 30 seconds.

m	s	
⑥ ① 7 2	⑩ 15	15 seconds – 30 seconds We borrow 1 minute = 60 seconds
– 5 8	30	So, 60 + 15 = 75 seconds 75 – 30 = 45 seconds
1 3	45	Thus, 13 minutes 45 seconds

EXERCISE 5.5

A. Add.

- 30 minutes 38 seconds and 20 minutes 42 seconds.
- 47 minutes 25 seconds and 19 minutes 49 seconds.
- 4 hours 40 minutes and 3 hours 57 minutes.
- 3 hours 35 minutes 26 seconds and 2 hours 40 minutes 50 seconds.
- 2 hours 55 minutes 45 seconds and 3 hours 48 minutes 44 seconds.

B. Subtract.

- (1) 38 minutes 39 seconds from 62 minutes 20 seconds.
- (2) 44 minutes 25 seconds from 50 minutes.
- (3) 2 hours 58 minutes from 3 hours 5 minutes
- (4) 4 hours 32 minutes from 7 hours.
- (5) 3 hours 45 minutes 50 seconds from 5 hours 30 minutes 40 seconds.

5.2.3 Convert years to months, months to days, weeks to days and vice versa

We already know that:

24 hours	=	1 day
7 days	=	1 week
1 month	=	30 days
12 months	=	1 year
365 days	=	1 year

A. Convert years to month and months to year

To convert years into months, we multiply the number of years by 12.

Example.

- (a) Convert 11 years to months
- (b) Convert 5 years 10 months to months

Solution:

- (a) 11 years = (11×12) months = 132 months
- (b) 5 years 10 months = 5 years + 10 months
= (5×12) months + 10 months
= 60 months + 10 months
= 70 months

Teacher's Note

Teacher should explain the students method of conversion of years to months, months to days, week to days and vice versa and give them more task.

B. Convert months to years

To convert months to years, we divide the number of months by 12.

Example.

- (a) Convert 48 months to years
- (b) Convert 81 months to years and months

Solution:

- (a) 48 months = $(48 \div 12)$ years = 4 years
- (b) 81 months = $(81 \div 12)$ years = $(72 \div 12)$ years + 9 months
= 6 years 9 months

C. Convert months to days

To convert months to days, we multiply the number of months by 30.

Example .

- (a) Convert 18 months to days
- (b) Convert 13 months 25 days to days

Solution:

- (a) 18 months = 18×30 days = 540 days
- (b) 13 months 25 days = 13 months + 25 days
= (13×30) days + 25 days
= 390 days + 25 days = 415 days

D. Convert days to months

To convert days to months, we divide the number of days by 30.

Example.

- (a) Convert 150 days to months
- (b) Convert 244 days to months and days

Solution:

- (a) 150 days = $(150 \text{ days} \div 30)$ months = 5 months
- (b) 244 days = $(244 \div 30)$ months
= $(240 \div 30)$ months + 4 days
= 8 months 4 days

E. Convert weeks to days

To convert weeks to days, we multiply the number of weeks by 7.

Example.

- (a) Convert 4 weeks to days
 (b) Convert 14 weeks 2 days to days

Solution:

- (a) $4 \text{ weeks} = 4 \times 7 \text{ days} = 28 \text{ days}$
 (b) $14 \text{ weeks } 2 \text{ days} = 14 \text{ weeks} + 2 \text{ days}$
 $= (14 \times 7) \text{ days} + 2 \text{ days}$
 $= 98 \text{ days} + 2 \text{ days} = 100 \text{ days}$

F. Convert days to weeks

To convert days into weeks, we divide the number of days by 7.

Example.

- (a) Convert 98 days to weeks
 (b) Convert 125 days to weeks and days

Solution:

- (a) $98 \text{ days} = (98 \div 7) \text{ weeks} = 14 \text{ week}$
 (b) $125 \text{ days} = (125 \div 7) \text{ weeks}$
 (Divide $125 \div 7$, we get 17 weeks and remainder 6 days)
 $= 17 \text{ weeks } 6 \text{ days}$

$$\begin{array}{r} 17 \\ 7 \overline{) 125} \\ \underline{- 7} \\ 55 \\ \underline{- 49} \\ 6 \end{array}$$

EXERCISE 5.6**Convert:**

- (1) 46 days into weeks and days.
- (2) 80 days into weeks and days.
- (3) 213 days into weeks and days.
- (4) 2343 days into weeks and days.
- (5) 450 days into months and days.

- (6) 800 days into months and days.
- (7) 710 days into months and days.
- (8) 650 days into months and days.
- (9) 66 months into years and months.
- (10) 82 months into years and months.
- (11) 49 months into years and months.
- (12) 244 months into years and months.

5.2.4 Solve real life problems involving conversion, addition and subtraction of units of time.

Example 1. Sana is 12 years 8 months old and her sister Saba is 6 years 10 months old. How much older is Sana than her sister?

Solution: We have to subtract the ages of Saba and Sana.

	Years	Months
Sana's age	¹¹ 12	¹² 8
Saba's age	- 6	10
	5	10

First we borrow 1 year = 12 months by adding 12 months + 8 months = 20 months
Now 20 - 10 months
Write 10 below months column.

Sana is 5 years and 10 months older than her sister Saba.

Example 2. The school opens at 7:30 a.m and closes at 1:15 p.m. How much time does the school remain open?

Solution:

Duration from 7:30 a.m to 12:00 noon = 4 hours 30 minutes

Duration from 12:00 noon to 1:15 p.m = 1 hour 15 minutes

Now,

	Hours	Minutes
	4	30
	+ 1	15
	5	45

School duration is 5 Hours and 45 minutes.

EXERCISE 5.7

1. Umair plays for 1 hour 15 minutes and Ali plays for 55 minutes. Who plays for shorter period and how much?
2. Fatima takes 1 hour 20 minutes to complete her Mathematics homework and 50 minutes to complete her English homework. How much total time she takes to complete the homework of both subjects?
3. School closes for summer vacations for a period of 2 months and 5 days, while the other holidays are for 27 days. How many days school will remain closed?
4. At a library Hassan reads a newspaper for 36 minutes and then reads a magazine for 1 hour 50 minutes. How long does he spend in reading?
5. Adil takes 2 hours 9 minutes to complete a video game, while Ali takes 55 minutes to complete the same game. How much more time Adil takes as compared to Ali?

5.3 TEMPERATURE

5.3.1 Recognize scales of temperature in Fahrenheit and Celsius

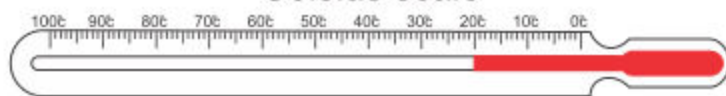
The instrument used to measure temperature is called thermometer.

There are two types of scales used in thermometers.

1. Celsius scale

2. Fahrenheit scale

Celsius scale



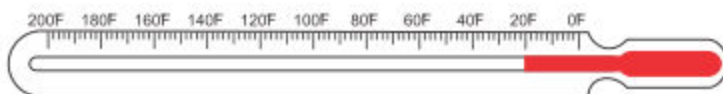
Celsius scale

The freezing point of water is zero degrees Celsius (0°C) and the boiling point of water is 100°C on the Celsius scale. The scale is divided into 100 equal parts. This scale is named after the Swedish Astronomer Celsius scale and is used in most parts of the world.

Teacher's Note

Teacher should explain the concept of temperature and give knowledge about different scales of temperature.

Fahrenheit scale

**Fahrenheit scale**

The freezing point of water is 32 degrees Fahrenheit (32°F) and the boiling point of water is 212 °F on the Fahrenheit scale, the scale is divided into 180 equal parts. This scale is named after the German scientist G. Fahrenheit.

(i) Converting temperature from Fahrenheit scale to Celsius scale

To convert temperature from Fahrenheit to Celsius scale, we subtract 32 and then multiply the difference by $\frac{5}{9}$.

Example: Change 59°F to the Celsius scale.

Solution:

Step 1: Subtraction of 32 from 59 gives

$$59 - 32 = 27$$

Step 2: Multiplication of 27 by $\frac{5}{9}$ gives

$$\overset{3}{\cancel{27}} \times \frac{5}{\cancel{9}} = 3 \times 5 = 15$$

Thus, 59°F = 15°C.

(ii) Converting from Celsius scale to Fahrenheit scale

To convert the temperature from Celsius scale to the Fahrenheit scale, we multiply given temperature by $\frac{9}{5}$ and add 32 to the product.

Example: Change 35°C to the Fahrenheit scale.

Solution:

Step 1: Multiplication of given temperature by $\frac{9}{5}$ gives

$$35 \times \frac{9}{5} = 7 \times 9 = 63$$

Step 2: Addition of 32 and 63 gives

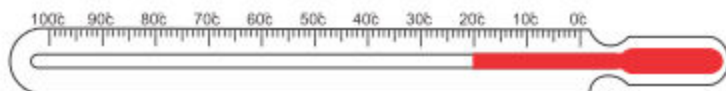
$$63 + 32 = 95$$

Thus, $35^{\circ}\text{C} = 95^{\circ}\text{F}$.

EXERCISE 5.8

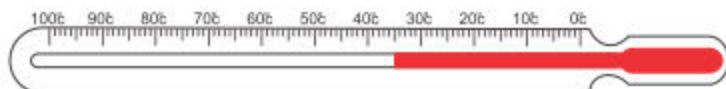
A. Write the readings from following thermometer.

(1)



____ $^{\circ}\text{C}$

(2)



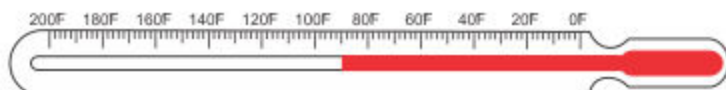
____ $^{\circ}\text{C}$

(3)



____ $^{\circ}\text{F}$

(4)



____ $^{\circ}\text{F}$

B. Change to the Celsius scale:

(1) 41°F

(2) 77°F

(3) 95°F

(4) 68°F

(5) 203°F

(6) 230°F

(7) 257°F

(8) 275°F

(9) 122°F

(10) 167°F

(11) 230°F

(12) 248°F

C. Change to the Fahrenheit scale:

- (1) 30°C (2) 45°C (3) 85°C (4) 55°C (5) 90°C
(6) 10°C (7) 20°C (8) 60°C (9) 80°C (10) 110°C

5.3.2 Solve real life problems involving conversion, addition and subtraction of units of temperature**Example:**

In the month of May on one day the maximum temperature in Larkana was 113°F. What was the temperature in degree Celsius?

Solution: Temperature in Larkana was 113°F

Subtracting 32 from 113 gives: $113 - 32 = 81$

Multiplication of 81 by $\frac{5}{9}$ gives: $81 \times \frac{5}{9} = \frac{81 \times 5}{9} = 45$

Thus, temperature in Celsius was 45°C

EXERCISE 5.9

- (1) Day the maximum temperature on a day in Hyderabad was 35°C. What was the temperature in degree Fahrenheit?
- (2) During summer on one day the temperature in Jacobabad was 113°F where as the temperature in Hyderabad was 40°C. Which city had more temperature and how much?
- (3) Ali was suffering from fever. His temperature at noon was 102°F. Convert the said temperature into degree Celsius.
- (4) In summer on one day the maximum temperature in Sukkur was 104°F, where as in Karachi it was 35°C. What was the difference in the temperatures of both the cities in degree Fahrenheit?
- (5) Day the maximum temperature on a day in Sehwan Sharif was 86°F. The minimum temperature on that day was 20°C. What was the difference in the both temperatures in degree Fahrenheit?

REVIEW EXERCISE 5

1. Convert:

- (i) 28km 648m to m (ii) 48m 97cm to cm
(iii) 76cm 9mm to mm (iv) 6m 75cm 8mm to mm

2. Add:

- (i) 38 minutes 42 seconds and 25 minutes 55 seconds
(ii) 20 hours 30 minutes and 16 hours 43 minutes
(iii) 4m 70cm 9mm and 3m 80cm 5mm

3. Subtract:

- (i) 22km 895m from 67km 472m
(ii) 2m 85cm 8mm from 4m 13cm 2mm
(iii) 39 hours 49 minutes from 76 hours 32 minutes

4. Choose the correct answer:

- (i) 5 km =
(a) 500 m (b) 5000 m (c) 50 m (d) 555 m
- (ii) 3 metres =
(a) 30 cm (b) 300 cm (c) 3000 cm (d) 30000 cm
- (iii) 6 weeks =
(a) 40 days (b) 35 days (c) 30 days (d) 42 days
- (iv) $59^{\circ}\text{F} = \text{ }^{\circ}\text{C}$
(a) 15 (b) 25 (c) 30 (d) 35

5. What will be the time?

- (i) 50 minutes earlier than 13 : 25
(ii) 45 minutes later than 9 : 45

6. How much time?

- (i) 5 : 30 to 6 : 15 (ii) 2 : 20 to 3 : 25